PyGameMaker — Tutorial 8: Lunar Lander

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Lunar Lander: Land on the Moon** (4 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A lunar lander game: your spacecraft falls slowly under the Moon's weak gravity. You use the arrow keys to thrust and steer, and you must land on the green pad. Hit the rocky ground and you crash.

Phase 1: A Flying Lander

- ☐ **1.** Create two sprites (32×32): spr_lander and spr_ground.
- ☐ **2.** Create obj_ground (sprite spr_ground, **Solid**) and obj_lander (sprite spr_lander).
- ☐ **3.** In obj_lander, **Create** event: *Set gravity* direction **270**, strength **0.05**. (The Moon's gravity is about a sixth of Earth's, so the value is small.)
- ☐ **4. Keyboard: Up Arrow (held)** with *Set vertical speed -2*.
- ☐ **5. Keyboard: Left Arrow (held)** with *Set horizontal speed -2* and **Right Arrow (held)** with *2*.
- ☐ **6. Keyboard: No key** with *Set horizontal speed 0*. Do not reset the vertical speed, or gravity stops working.
- ☐ **7.** Create room_game (640×480, **black** background, Snap to Grid 32×32). Put ground along the bottom row and the lander near the top.

> **You should see:** press **F5**. The lander falls slowly. Hold **Up** to climb, **Left** and **Right** to steer. Let go of everything and it drifts down. (It passes through the ground for now: no collision event yet.)

Phase 2: Landing and Crashing

- ☐ **8.** Create spr_pad (a flat green rectangle) and obj_pad (sprite spr_pad, **Solid**).
- ☐ **9.** In obj_lander, add **Collision with obj_pad**: *Stop movement*, then *Set gravity* direction 270 strength **0**, then *Show message* "Landing successful!".
- ☐ **10.** Add **Collision with obj_ground**: *Show message* "Crashed!", then *Restart room*.
- ☐ **11.** Change the room: make the ground uneven with hills, leave a gap in the bottom row, and put obj_pad in the gap (one or two blocks).

> **You should see:** press **F5**. Touch the pad and you get "Landing successful!" once. Touch the ground and you get "Crashed!" and the level restarts. Use short taps of **Up** to control your fall.

Phase 3: Game Controller

- ☐ **12.** Create obj_game_controller (no sprite). **Create** event: *Set score to 0*.

- [ ] **13. Draw** event: first *Set draw color* to white (#ffffff), then *Draw text* "Lunar Lander" at x 10, y 10 and *Draw text* "Land on the green pad!" at x 10, y 30.
- [ ] **14.** Place obj_game_controller anywhere in the room.

> **You should see:** press **F5**. The title and the instructions are shown at the top left in white.

Stuck?

What is wrong	Most likely cause	What to do
The lander does not fall	The gravity action is missing from **Create**, or **No key** uses *Stop movement*	Add gravity; make **No key** *Set horizontal speed 0*
The lander falls very fast	The gravity strength is too big (for example 5 instead of 0.05)	Set the strength to 0.05
The lander goes through the ground	No **Collision with obj_ground**, or obj_ground is not Solid	Add the event; tick **Solid**
"Landing successful!" keeps popping up	Gravity is not switched off after landing	Add *Set gravity* strength 0 to the pad event
I crash every time, even on the pad	The pad is missing, or too small or too high to reach	Check obj_pad is in the room and Solid; use one or two blocks
The lander sticks to the ground after a crash	*Restart room* is missing	Add it after *Show message*
The text does not show	The text is black on the black room	Add *Set draw color* white before the text
The lander is too hard to control	Long presses of Up make it climb fast	Tap Up in short bursts

Challenges

- **Try this (5 minutes):** change the gravity or the thrust and see how the game feels.
- **Push further:** make the pad smaller, or build a second, harder level in another room.
- **Invent:** add a fuel counter that goes down while you thrust and stops the thrust at 0, or make a landing that is too fast count as a crash.

Vocabulary

Term	What it means
Thrust	A push that fights gravity (here, setting the vertical speed to -2)
Gravity	A force that pulls down a little more every step
Vertical / horizontal speed	How fast the lander moves up-down / left-right
Landing pad	The only place where touching down is a success
HUD	Text on the screen that gives the player information

Check Yourself

Write your answers in the notes below.

1. Why is the gravity value 0.05 and not 0.5 as in the platformer?
2. Why does **No key** reset only the horizontal speed?
3. Why must gravity be switched off when the lander lands on the pad?

My Notes